1.) Clearing Messy Data (Algorithm)

step-1: Start the program
step-2: Import the required libraries: pandas, seaborn
step-3: load the Titanic dataset using seaborn
step-4: select the required columns: survived, pclass, sex, age, fare, embarked
step-5: check the missing values in each column.
step-6: Handle missing values in age column:
step-7: Replace missing values in embarked column as 'S'
step-8: convert categorical columns into numerical columns.
step-9: Display the processed dataset.
step-10: check whether any missing values are present
step-11: stop

code flow:

* Import pandas as pd , import seaborn as sns ; these are the tools.
* pandas gives information about tables and seaborn gives us ready made datasets.
* We have to load the dataset into table called data.
* data must contain useful columns called survived, pclass, sex, age, fare, embarked.
* Check and fill missing values using median.
* Convert categorical values into numerical values using get_dummies().
* Display the cleaned dataset

2. Train & Test fairly CKNN on Iris Dataset.

Algorithm:

* Start
* Load dataset : load the Iris flower datasets containing measurements and special lables.
* separate and extract : feature measurements (x) and target species categories (y).
* Divide the dataset into an 80% training set and a 20% test set to evaluate the model fairly.
* ~~Creat~~ Initialize the K-NN classifier with K=5.
* Train the model using training set (Xtrain, Ytrain).
* score the model accuracy on the unseen test set (Xtest, Ytest)
* ~~Pass~~ Predict the new flower species.
* Stop the process.

Code flow:

* Import the required libraries
* Load the Iris dataset
* Split the dataset into training and testing.
* test_size = 0.2 means 80% training and 20% testing data.
* If changed to 0.3, it becomes 70% training & 30% testing.
* Apply KNN algorithm
* Calculate & display the accuracy.

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3. Predict A Number (Linear Regression on Auto MPG Dataset)

Algorithm:

* Start the program
* ~~Load~~ Read Auto MPG dataset using seaborn and remove rows with missing values.
* Automobile specifications is input (x) and fuel efficiency (mpg) is target continuous variable (y).
* Divide dataset into 80% training & 20% set
* Initialize Linear Regression model.
* fit the model to learn mathematical relation b/w car feature and mpg.
* Compute R^2 score.
* Input new vehicle parameters to predicts its mpg.

codeflow:

* Import pandas, Numpy and Scikit-learn.
* load the Auto MPG dataset.
* Select input and output features.
* Train the linear Regression model.
* fit() is used to train the model.
* If removed, the model will not be trained.
* Predict mileage and display the result.

4. Predict yes/no (KNN & SVM on social network Ads)

Algorithm:

* Start the process.
* Read social_network_Ads.csv containing user age, estimated salary and purchase outcome.
* Use Age and Estimated Salary as inputs (x) and purchased (0 or 1) as the target (y).
* split the data into 75% training & 25% splits testing.
* Input values using standard Scaler so high values donot dominate low values.
* create and train KNN and SVM classifier.
* Calculate and compare accuracy scores for both models.
* Transform a new user's age and salary & predict.
* Stop.

Code flow:

Exp 4:

* Import the required libraries
* load the Social Network Ads dataset.
* separate input and output features
* Apply K-Means Clustering.
* predict() is used to predict the output.
* It is called after training the model
* ~~Display customer~~ groups vs
* Apply KNN & SVM algorithms and display accuracy

5. Find Groups CK-Means on Mall customers

Algorithm:

* Start the program.
* Read Mall-customers.csv using pandas.
* Use Annual Income and spending score as input
* set n_clusters =5 to divide into 5 groups.
* Apply K-Means to find customer groups.
* Store the cluster number for each customer.
* Give the income and spending score of new customer and identify its group.
* plot customers as points and display the 5 cluster centroids.

Code flow:

Exp 5:

* Import pandas, matplotlib & K-Means.
* Load the Mall Customers dataset.
* Select annual Income and spending score.
* Apply K-Means clustering.
* accuracy_score() is used to calculate model accuracy
* Higher accuracy means more correct predictions

6. Groups as a Tree (Hierarchical Clustering on Mall Customers)

Algorithm:

* start
* Read Mall_Customers.csv using pandas:
* use Annual Income and spending score as input.
* Set n-clusters = 5 using agglomerative bottom-up hierchial clustering.
* Cluster data points by merging nearest pairs.
* Compute mean feature values for each cluster.
* determine nearest cluster center using Euclidean distance to classify new data point.

Code flow:

* import the required libraries.
* Load the Mall Customers dataset
* select the required feature.
* Apply Hierarchical Clustering.
* draw and display the dendrogram.

7. A forest of Trees (Random forest on Breast Cancer)

Algorithm:

* Start
* Load the breast cancer dataset.
* Split 80% to training and 20% to testing.
* Build and fit a single DecisionTreeclassifier for comparison
* fit an ensemble Randomforest Classifier with 100 decision tree.
* Calculate accuracy for both value.
* Evaluate new sample to predict whether tumor is benign or malignant

Codeflow:

* import the required libraries.
* Load the Breast Cancer dataset.
* split the data into training and testing sets.
* Apply the Random Forest algorithm
* predict the result and Calculate accuracy.

8: Boosting (Gradient Boosting on Heart Disease)

Algorithm:

* load d heart.csv dataset.
* set input(x) as clinical parameters and heartdisease as target(y).
* Split 80% to train & 20% to test.
* Initialize GradientBoostingClassifier model.
* Train the tree models.
* Score test set predictions to measure diagnosis accuracy on unseen records
* predict whether new patient's results indicate heartdisease.

codeflow:

Exp 8:

* Import the required libraries.
* Load the Heart Disease dataset.
* separate input and output features.
* Train the Boosting model.
* Predict the result and display accuracy.

9. Learn by Reward (Q-learning on frozenlake)

Algorithm:

* Initialize Frozenlake-v1 using Gymnasium.
* Construct a matrix of zeros initialised to size (states x actions) =16x4
* Iterate through multiple practice episodes.
* Identify max Q-values for target state inputs.

codeflow:

* Import the Gymnasium and Numpy libraries
* Create the FrozenLake game environment.
* Initialize the Q-table.
* Train the agent using rewards and penalties.
* Display the learned result

10. A Neural Network (Handwritten Digits)

Algorithm:

* load Handwritten dataset.
* Split 80% to train & 20% to test.
* Build Neural network Architecture.
* compile model using adam and sparse_categorical_cross entropy.
* train across 10 parameters.
* predict class index for unseen pixel matrix input.

Code flow:

Exp 10:

* Import Tensorflow and scikit-learn libraries.
* Load the Handwritten Digits dataset.
* Split the data into training and testing sets.
* Build and train the Neural Network model.
* Predict handwritten digits and display accuracy.